

## NEXO VEST, Denmark

WMO: 061970

Lat: 55.0558N

Lon: 15.0953E

Elev: 82

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 21.6       | 24.4 | 14.1                          | 11.3 | 24.2 | 17.0 | 13.0 | 26.0 | 36.9                  | 39.3 | 34.4 | 39.3 | 11.5                  | 120  | 0.659 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.2                   | 76.2            | 66.5 | 73.0 | 64.9 | 70.8 | 63.6 | 68.7                | 73.7 | 66.5 | 70.8 | 64.9 | 69.0 | 9.7                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.7                            | 98.7 | 71.3 | 64.8 | 92.1 | 69.0 | 63.2 | 86.9 | 67.1 | 33.0          | 73.9 | 31.2 | 70.8 | 29.9 | 68.9 | 75.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 29.2              | 25.7 | 23.2 | 18.7                       | 82.4 | 4.1                | 3.4 | 15.7                                               | 84.8 | 13.3       | 86.8 | 11.0       | 88.7 | 8.0        | 91.2 |    |    |
|                   |      |      | 17.5                       | 71.0 | 4.2                | 2.6 | 14.5                                               | 72.9 | 12.0       | 74.4 | 9.6        | 75.9 | 6.6        | 77.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 47.9   | 34.0  | 34.2  | 37.6  | 43.6  | 51.2  | 57.6  | 63.3  | 63.4  | 57.8  | 49.7  | 43.4  | 38.6 |
|                                                                               | DBStd       | 11.42  | 5.40  | 5.09  | 5.12  | 5.05  | 5.28  | 4.35  | 4.32  | 4.06  | 4.13  | 4.66  | 4.73  | 4.76 |
|                                                                               | HDD50       | 2196   | 496   | 443   | 384   | 202   | 50    | 2     | 0     | 0     | 2     | 62    | 201   | 354  |
|                                                                               | HDD65       | 6298   | 961   | 863   | 849   | 643   | 429   | 227   | 87    | 79    | 219   | 474   | 648   | 819  |
|                                                                               | CDD50       | 1441   | 0     | 0     | 0     | 9     | 86    | 229   | 411   | 415   | 235   | 53    | 3     | 0    |
|                                                                               | CDD65       | 69     | 0     | 0     | 0     | 0     | 0     | 4     | 34    | 29    | 2     | 0     | 0     | 0    |
|                                                                               | CDH74       | 195    | 0     | 0     | 0     | 0     | 0     | 18    | 106   | 70    | 1     | 0     | 0     | 0    |
|                                                                               | CDH80       | 14     | 0     | 0     | 0     | 0     | 0     | 1     | 6     | 7     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 11.3   | 13.6  | 12.9  | 12.2  | 10.4  | 9.5   | 9.3   | 8.8   | 9.6   | 11.2  | 12.0  | 12.6  | 13.6 |
| Precipitation                                                                 | PrecAvg     | 22.2   | 1.6   | 1.2   | 1.4   | 1.6   | 1.6   | 1.4   | 2.1   | 2.2   | 2.4   | 2.1   | 2.6   | 2.2  |
|                                                                               | PrecMax     | 27.8   | 3.3   | 2.4   | 3.1   | 4.4   | 3.1   | 3.5   | 3.8   | 5.9   | 4.0   | 5.7   | 4.7   | 4.5  |
|                                                                               | PrecMin     | 17.7   | 0.4   | 0.4   | 0.3   | 0.5   | 0.4   | 0.5   | 0.3   | 0.3   | 0.8   | 0.7   | 0.6   | 0.6  |
|                                                                               | PrecStd     | 3.0    | 0.8   | 0.6   | 0.7   | 0.9   | 0.8   | 0.8   | 1.2   | 1.6   | 0.9   | 1.4   | 1.3   | 1.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 45.7  | 47.4  | 54.1  | 62.3  | 68.8  | 76.7  | 80.6  | 80.2  | 72.1  | 62.1  | 54.8  | 48.7 |
|                                                                               |             | MCWB   | 44.4  | 44.2  | 48.2  | 52.3  | 59.7  | 67.0  | 68.2  | 67.7  | 64.3  | 57.7  | 52.8  | 47.3 |
|                                                                               | 2%          | DB     | 44.0  | 44.1  | 48.9  | 57.7  | 65.4  | 70.6  | 76.9  | 75.4  | 68.2  | 59.4  | 52.8  | 47.3 |
|                                                                               |             | MCWB   | 42.8  | 41.7  | 44.3  | 50.3  | 58.4  | 62.7  | 66.5  | 66.6  | 62.0  | 56.6  | 51.1  | 45.9 |
|                                                                               | 5%          | DB     | 42.5  | 42.4  | 46.2  | 54.0  | 62.6  | 68.0  | 74.0  | 72.4  | 65.7  | 57.8  | 51.1  | 46.1 |
|                                                                               |             | MCWB   | 41.0  | 40.7  | 42.8  | 48.1  | 56.5  | 60.9  | 65.3  | 64.8  | 60.3  | 55.2  | 49.7  | 44.7 |
|                                                                               | 10%         | DB     | 41.3  | 41.1  | 44.1  | 51.1  | 60.1  | 65.8  | 71.5  | 70.3  | 64.0  | 56.2  | 49.6  | 44.8 |
|                                                                               |             | MCWB   | 40.0  | 39.6  | 41.5  | 46.5  | 54.8  | 59.5  | 64.3  | 63.3  | 59.5  | 53.8  | 47.9  | 43.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.8  | 44.5  | 48.3  | 54.3  | 62.0  | 68.3  | 71.7  | 71.0  | 66.1  | 59.2  | 53.7  | 48.1 |
|                                                                               |             | MCDB   | 45.2  | 45.9  | 51.8  | 59.4  | 66.7  | 75.1  | 76.5  | 76.6  | 69.2  | 60.9  | 54.7  | 48.4 |
|                                                                               | 2%          | WB     | 43.0  | 42.2  | 45.3  | 51.2  | 59.8  | 64.8  | 69.2  | 68.4  | 63.1  | 57.3  | 51.7  | 46.5 |
|                                                                               |             | MCDB   | 43.7  | 43.7  | 47.7  | 55.9  | 63.9  | 68.6  | 74.4  | 72.5  | 65.8  | 58.8  | 52.4  | 47.0 |
|                                                                               | 5%          | WB     | 41.4  | 41.0  | 43.4  | 49.0  | 57.6  | 62.5  | 67.2  | 66.2  | 61.9  | 55.7  | 49.9  | 45.1 |
|                                                                               |             | MCDB   | 42.2  | 42.4  | 45.4  | 53.1  | 61.4  | 66.3  | 71.9  | 70.5  | 64.7  | 57.4  | 51.2  | 45.8 |
|                                                                               | 10%         | WB     | 40.1  | 39.6  | 41.9  | 46.9  | 55.4  | 60.6  | 65.4  | 64.6  | 60.6  | 54.3  | 48.1  | 43.5 |
|                                                                               |             | MCDB   | 41.1  | 40.9  | 43.7  | 50.6  | 59.0  | 64.5  | 69.5  | 68.8  | 63.3  | 56.0  | 49.4  | 44.5 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 5.1   | 5.4   | 7.4   | 10.1  | 12.0  | 12.7  | 12.4  | 11.2  | 8.5   | 6.8   | 5.5   | 5.1  |
|                                                                               |             | MCDBR  | 5.4   | 7.2   | 10.9  | 14.9  | 15.5  | 15.4  | 15.7  | 13.6  | 10.2  | 7.8   | 5.6   | 5.4  |
|                                                                               |             | MCWBR  | 5.8   | 5.8   | 8.0   | 9.9   | 10.1  | 9.2   | 8.0   | 7.2   | 6.2   | 6.2   | 5.4   | 5.7  |
|                                                                               | 5% WB       | MCDBR  | 5.4   | 7.0   | 10.1  | 13.2  | 14.0  | 14.3  | 13.7  | 12.9  | 8.8   | 7.3   | 5.7   | 5.6  |
|                                                                               |             | MCWBR  | 5.8   | 5.7   | 8.0   | 9.6   | 9.7   | 9.6   | 8.1   | 8.1   | 6.2   | 6.0   | 5.6   | 6.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.319  | 0.343 | 0.365 | 0.398 | 0.396 | 0.388 | 0.412 | 0.413 | 0.379 | 0.363 | 0.344 | 0.314 |      |
|                                                                               | taud        | 2.310  | 2.331 | 2.313 | 2.286 | 2.366 | 2.416 | 2.366 | 2.365 | 2.433 | 2.395 | 2.330 | 2.298 |      |
|                                                                               | Ebn at Noon | 192    | 231   | 254   | 261   | 269   | 272   | 263   | 254   | 248   | 221   | 180   | 166   |      |
|                                                                               | Edh at Noon | 18     | 24    | 31    | 36    | 36    | 35    | 36    | 34    | 28    | 23    | 17    | 15    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 134    | 333   | 772   | 1368  | 1792  | 1920  | 1748  | 1412  | 1000  | 482   | 187   | 95    |      |
|                                                                               | RadStd      | 23     | 59    | 68    | 138   | 181   | 120   | 192   | 107   | 98    | 76    | 30    | 12    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (49 neighbors) | +0.72 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -274        | +106  | N/A   |       |

Nomenclature: See separate page